

- Description

This module manages a smart refrigerator door open duration monitoring system, ensuring that:

1. The system continuously monitors the door open time (in seconds).
2. If the door remains open longer than a predefined maximum duration, an alert is triggered to prompt the user to close the door and save energy.
3. If the door open time is within the acceptable range, no action is taken.

This system helps reduce energy loss and maintain optimal cooling performance by ensuring the refrigerator door is not left open for too long.

- Specification

module SmartRefrigeratorDoorControl

var currentOpenTime, maxOpenTime: real

process updateDoorOpenTime(newOpenTime: real) returns (currentOpenTime: real)

functional scenario form:

currentOpenTime > maxOpenTime && output "ALERT: Door open time (" + currentOpenTime + " s) exceeds maximum allowed (" + maxOpenTime + " s). Please close the door to save energy.", Alert triggered for excessive door open time.

||

currentOpenTime <= maxOpenTime && output "Door open time (" + currentOpenTime + " s) is within safe limits.", No action required; door open time is acceptable.

end process

end module

- Java Code

```
package code;
public class c105_SmartRefrigeratorDoor {
    private double maxOpenTime;
    private double currentOpenTime;
    public c105_SmartRefrigeratorDoor(double maxOpenTime) {
        this.maxOpenTime = maxOpenTime;
        this.currentOpenTime = 0.0;
    }

    public void updateDoorOpenTime(double newOpenTime) {
        currentOpenTime = newOpenTime;
        System.out.printf("Current Door Open Time: %.2f seconds\n", currentOpenTime);
        if (currentOpenTime > maxOpenTime) {
            System.out.println("ALERT: Door has been open too long! Please close the door to save energy.");
        } else {
```

```

        System.out.println("Door open time is within safe limits.");
    }
}

public double getCurrentOpenTime() {
    return currentOpenTime;
}

public static void main(String[] args) {
    c105_SmartRefrigeratorDoor doorMonitor = new c105_SmartRefrigerato
rDoor(30.0);
    // testcase-VT:
    //     doorMonitor.updateDoorOpenTime(20.0);
    //     doorMonitor.updateDoorOpenTime(35.0);
    //     doorMonitor.updateDoorOpenTime(25.0);

    // testcase-FT:
    //     doorMonitor.updateDoorOpenTime(1.6349922841177022E296);
    //     doorMonitor.updateDoorOpenTime(-1.7976931344437577E308);
    //     doorMonitor.updateDoorOpenTime(1.9490628022799996E289);

    // testcase-Z3:
    //     doorMonitor.updateDoorOpenTime(20.0);
    //     doorMonitor.updateDoorOpenTime(35.0);
    //     doorMonitor.updateDoorOpenTime(25.0);

    // testcase-UVT:
    doorMonitor.updateDoorOpenTime(20.0);
    doorMonitor.updateDoorOpenTime(35.0);
    doorMonitor.updateDoorOpenTime(25.0);
}
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c105_SmartRefrigeratorDoor.java	100.0 %	2	0	2
▼ c105_SmartRefrigeratorDoor	100.0 %	2	0	2
● updateDoorOpenTime(double)	100.0 %	2	0	2
● c105_SmartRefrigeratorDoor(do		0	0	0
● main(String[])		0	0	0

3. Vibration testing instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	53	21,870	21,923
▼ src	0.2 %	53	21,870	21,923
▼ code	0.2 %	53	21,870	21,923
▼ c105_SmartRefrigeratorDoor.java	94.6 %	53	3	56
▼ c105_SmartRefrigeratorDoor	94.6 %	53	3	56
● updateDoorOpenTime(double)	100.0 %	29	0	29
● c105_SmartRefrigeratorDoor(do	100.0 %	9	0	9
● main(String[])	100.0 %	15	0	15

4. Fuzz testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c105_SmartRefrigeratorDoor.java	100.0 %	2	0	2
▼ c105_SmartRefrigeratorDoor	100.0 %	2	0	2
● updateDoorOpenTime(double)	100.0 %	2	0	2
● c105_SmartRefrigeratorDoor(do		0	0	0
● main(String[])		0	0	0

5. Fuzz testing instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	53	21,870	21,923
▼ src	0.2 %	53	21,870	21,923
▼ code	0.2 %	53	21,870	21,923
▼ c105_SmartRefrigeratorDoor.java	94.6 %	53	3	56
▼ c105_SmartRefrigeratorDoor	94.6 %	53	3	56
● updateDoorOpenTime(double)	100.0 %	29	0	29
● c105_SmartRefrigeratorDoor(do	100.0 %	9	0	9
● main(String[])	100.0 %	15	0	15

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c105_SmartRefrigeratorDoor.java	100.0 %	2	0	2
▼ c105_SmartRefrigeratorDoor	100.0 %	2	0	2
● updateDoorOpenTime(double)	100.0 %	2	0	2
● c105_SmartRefrigeratorDoor(do		0	0	0
● main(String[])		0	0	0

7. Z3 instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	53	21,870	21,923
▼ src	0.2 %	53	21,870	21,923
▼ code	0.2 %	53	21,870	21,923
▼ c105_SmartRefrigeratorDoor.java	94.6 %	53	3	56
▼ c105_SmartRefrigeratorDoor	94.6 %	53	3	56
● updateDoorOpenTime(double)	100.0 %	29	0	29
● c105_SmartRefrigeratorDoor(do	100.0 %	9	0	9
● main(String[])	100.0 %	15	0	15

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c105_SmartRefrigeratorDoor.java	100.0 %	2	0	2
▼ c105_SmartRefrigeratorDoor	100.0 %	2	0	2
● updateDoorOpenTime(double)	100.0 %	2	0	2
● c105_SmartRefrigeratorDoor(do		0	0	0
● main(String[])		0	0	0

9. Unimproved V-method instruction coverage: 94.6%

Element		Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼  VibrationTestingTool		 0.2 %	53	21,870	21,923
▼  src		 0.2 %	53	21,870	21,923
▼  code		 0.2 %	53	21,870	21,923
▼  c105_SmartRefrigeratorDoor.java		 94.6 %	53	3	56
▼  c105_SmartRefrigeratorDoor		 94.6 %	53	3	56
 updateDoorOpenTime(double)		 100.0 %	29	0	29
 c105_SmartRefrigeratorDoor(do		 100.0 %	9	0	9
 main(String[])		 100.0 %	15	0	15